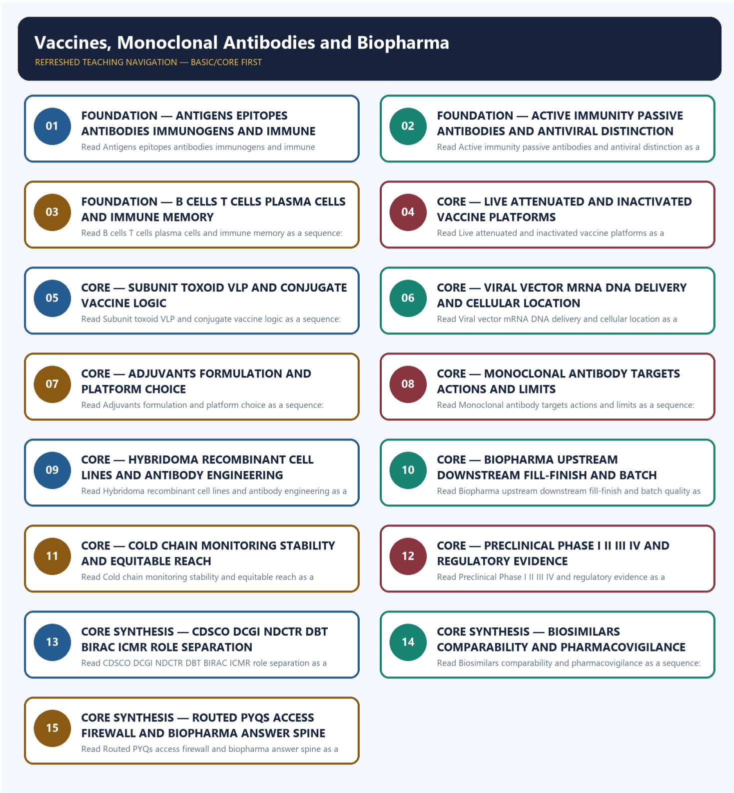
SOLVED PRACTICE WORKBOOK

Vaccines, Monoclonal Antibodies and Biopharma - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



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CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Vaccines, Monoclonal Antibodies and Biopharma - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Antigen-antibody boundary?

A. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. B. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. C. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. D. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication.

Answer: A. Explanation: An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Antigen-antibody boundary?

A. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. B. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. C. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. D. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory.

Answer: B. Explanation: An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Antigen-antibody boundary without changing its institution, unit or status?

A. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. B. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. C. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. D. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical.

Answer: C. Explanation: An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen,

antibody and pathogen are related but non-interchangeable terms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Antigen-antibody boundary?

A. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. B. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. C. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. D. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms.

Answer: D. Explanation: An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies Active-passive-antiviral boundary?

A. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. B. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. C. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. D. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection.

Answer: A. Explanation: Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of Active-passive-antiviral boundary?

A. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. B. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. C. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. D. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory.

Answer: B. Explanation: Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses Active-passive-antiviral boundary without changing its institution, unit or status?

A. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. B. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. C. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. D. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory.

Answer: C. Explanation: Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about Active-passive-antiviral boundary?

A. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. B. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. C. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. D. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication.

Answer: D. Explanation: Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Adaptive-immune mechanism?

A. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. B. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. C. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. D. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical.

Answer: A. Explanation: Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Adaptive-immune mechanism?

A. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. B. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. C. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. D. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration.

Answer: B. Explanation: Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Adaptive-immune mechanism without changing its institution, unit or status?

A. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. B. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. C. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. D. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects.

Answer: C. Explanation: Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Adaptive-immune mechanism?

A. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. B. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. C. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. D. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection.

Answer: D. Explanation: Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. The other options change the scientific category, institutional owner, measurement, mission rung

or operating status.

Q13. Which statement correctly identifies Whole-pathogen platform boundary?

A. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. B. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. C. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. D. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ.

Answer: A. Explanation: Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Whole-pathogen platform boundary?

A. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. B. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. C. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. D. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration.

Answer: B. Explanation: Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Whole-pathogen platform boundary without changing its institution, unit or status?

A. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. B. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. C. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. D. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration.

Answer: C. Explanation: Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Whole-pathogen platform boundary?

A. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. B. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. C. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. D. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical.

Answer: D. Explanation: Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Subunit-toxoid-VLP-conjugate boundary?

A. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. B. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. C. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. D. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects.

Answer: A. Explanation: Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Subunit-toxoid-VLP-conjugate boundary?

A. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. B. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. C. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. D. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects.

Answer: B. Explanation: Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Subunit-toxoid-VLP-conjugate boundary without changing its institution, unit or status?

A. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. B. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. C. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. D. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages.

Answer: C. Explanation: Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Subunit-toxoid-VLP-conjugate boundary?

A. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. B. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory.

Answer: D. Explanation: Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Vector-mRNA-DNA boundary?

A. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. B. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. C. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. D. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages.

Answer: A. Explanation: A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing

implications differ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Vector-mRNA-DNA boundary?

A. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. B. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects.

Answer: B. Explanation: A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Vector-mRNA-DNA boundary without changing its institution, unit or status?

A. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. B. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. C. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. D. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages.

Answer: C. Explanation: A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Vector-mRNA-DNA boundary?

A. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. B. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ.

Answer: D. Explanation: A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Adjuvant-and-formulation boundary?

A. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. B. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects.

Answer: A. Explanation: An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Adjuvant-and-formulation boundary?

A. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. B. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules.

Answer: B. Explanation: An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Adjuvant-and-formulation boundary without changing its institution, unit or status?

A. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. B. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. C. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. D. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs.

Answer: C. Explanation: An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an

adjuvant does not itself establish efficacy, safety or duration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Adjuvant-and-formulation boundary?

A. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. B. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. C. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. D. An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration.

Answer: D. Explanation: An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Monoclonal-antibody mechanism?

A. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. B. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages.

Answer: A. Explanation: A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Monoclonal-antibody mechanism?

A. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. B. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. C. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. D. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules.

Answer: B. Explanation: A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Monoclonal-antibody mechanism without changing its institution, unit or status?

A. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. B. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. C. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. D. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs.

Answer: C. Explanation: A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Monoclonal-antibody mechanism?

A. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. B. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. C. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. D. A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects.

Answer: D. Explanation: A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Hybridoma-recombinant-production boundary?

A. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. B. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules.

Answer: A. Explanation: Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Hybridoma-recombinant-production boundary?

A. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. B. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. C. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. D. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.

Answer: B. Explanation: Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Hybridoma-recombinant-production boundary without changing its institution, unit or status?

A. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. B. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. C. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. D. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details.

Answer: C. Explanation: Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Hybridoma-recombinant-production boundary?

A. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. B. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. C. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. D. Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages.

Answer: D. Explanation: Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Antibody-engineering boundary?

A. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. B. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. C. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. D. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.

Answer: A. Explanation: Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Antibody-engineering boundary?

A. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. B. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. C. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. D. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.

Answer: B. Explanation: Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Antibody-engineering boundary without changing its institution, unit or status?

A. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. B. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. C. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. D. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

Answer: C. Explanation: Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Antibody-engineering boundary?

A. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. B. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. C. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. D. Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness.

Answer: D. Explanation: Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Biopharma-manufacturing chain?

A. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. B. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. C. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. D. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.

Answer: A. Explanation: Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Biopharma-manufacturing chain?

A. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. B. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. C. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. D. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

Answer: B. Explanation: Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Biopharma-manufacturing chain without changing its institution, unit or status?

A. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. B. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. C. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. D. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

Answer: C. Explanation: Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Biopharma-manufacturing chain?

A. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. B. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. C. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. D. Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules.

Answer: D. Explanation: Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Quality-and-scale-up boundary?

A. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. B. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. C. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. D. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.

Answer: A. Explanation: Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Quality-and-scale-up boundary?

A. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. B. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. C. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. D. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

Answer: B. Explanation: Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Quality-and-scale-up boundary without changing its institution, unit or status?

A. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. B. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. C. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. D. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

Answer: C. Explanation: Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Quality-and-scale-up boundary?

A. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. B. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. C. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. D. Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs.

Answer: D. Explanation: Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies Cold-chain boundary?

A. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. B. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. C. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. D. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details.

Answer: A. Explanation: Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of Cold-chain boundary?

A. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. B. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. C. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. D. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

Answer: B. Explanation: Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses Cold-chain boundary without changing its institution, unit or status?

A. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. B. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. C. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. D. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence.

Answer: C. Explanation: Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q52. Which option avoids the standard UPSC close-option trap about Cold-chain boundary?

A. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. B. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. C. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. D. Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.

Answer: D. Explanation: Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies Clinical-development ladder?

A. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. B. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. C. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. D. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access.

Answer: A. Explanation: Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of Clinical-development ladder?

A. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. B. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. C. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. D. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence.

Answer: B. Explanation: Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses Clinical-development ladder without changing its institution, unit or status?

A. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. B. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. C. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. D. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence.

Answer: C. Explanation: Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about Clinical-development ladder?

A. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. B. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. C. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. D. Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details.

Answer: D. Explanation: Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and

safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies CDSCO-NDCTR boundary?

A. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. B. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. C. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. D. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence.

Answer: A. Explanation: CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of CDSCO-NDCTR boundary?

A. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. B. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. C. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. D. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access.

Answer: B. Explanation: CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses CDSCO-NDCTR boundary without changing its institution, unit or status?

A. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. B. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. C. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. D. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs;

each product claim needs its own dated authoritative source.

Answer: C. Explanation: CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about CDSCO-NDCTR boundary?

A. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. B. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. C. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. D. CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

Answer: D. Explanation: CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies DBT-BIRAC-ICMR boundary?

A. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. B. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. C. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. D. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine.

Answer: A. Explanation: DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of DBT-BIRAC-ICMR boundary?

A. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. B. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. C. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. D. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence.

Answer: B. Explanation: DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses DBT-BIRAC-ICMR boundary without changing its institution, unit or status?

A. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. B. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. C. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. D. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source.

Answer: C. Explanation: DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about DBT-BIRAC-ICMR boundary?

A. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. B. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. C. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. D. DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role.

Answer: D. Explanation: DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Mission-and-policy status boundary?

A. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. B. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. C. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. D. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine.

Answer: A. Explanation: Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Mission-and-policy status boundary?

A. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. B. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. C. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. D. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms.

Answer: B. Explanation: Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Mission-and-policy status boundary without changing its institution, unit or status?

A. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. B. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. C. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. D. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source.

Answer: C. Explanation: Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Mission-and-policy status boundary?

A. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. B. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. C. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. D. Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access.

Answer: D. Explanation: Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Biosimilar-generic boundary?

A. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. B. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. C. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. D. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source.

Answer: A. Explanation: A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Biosimilar-generic boundary?

A. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. B. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. C. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. D. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source.

Answer: B. Explanation: A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Biosimilar-generic boundary without changing its institution, unit or status?

A. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. B. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. C. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. D. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication.

Answer: C. Explanation: A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Biosimilar-generic boundary?

A. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. B. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. C. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. D. A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence.

Answer: D. Explanation: A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies Pharmacovigilance-and-AEFI boundary?

A. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. B. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. C. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. D. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms.

Answer: A. Explanation: Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of Pharmacovigilance-and-AEFI boundary?

A. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. B. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. C. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. D. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection.

Answer: B. Explanation: Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses Pharmacovigilance-and-AEFI boundary without changing its institution, unit or status?

A. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. B. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. C. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. D. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection.

Answer: C. Explanation: Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about Pharmacovigilance-and-AEFI boundary?

A. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. B. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. C. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. D. Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine.

Answer: D. Explanation: Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Efficacy-effectiveness-access firewall?

A. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. B. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. C. An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. D. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication.

Answer: A. Explanation: Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Efficacy-effectiveness-access firewall?

A. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. B. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. C. Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. D. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical.

Answer: B. Explanation: Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Efficacy-effectiveness-access firewall without changing its institution, unit or status?

A. Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. B. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. C. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. D. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory.

Answer: C. Explanation: Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Efficacy-effectiveness-access firewall?

A. A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. B. Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. C. Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. D. Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source.

Answer: D. Explanation: Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Three representative audited cards cover the 2020 PCV, 2021 recombinant-vector and 2022 COVID-platform objective demands; the 2018 biopharma and 2022 vaccine-development GS-III demands; and the 2025 monoclonal-antibody objective demand. No unavailable objective key is invented, and the available 2025 key is not recorded or inferred.

PYQ DEMAND CARD 1 - 2020, 2021 and 2022 Prelims GS-I

Demand: Assess pneumococcal conjugate-vaccine logic, recombinant-vector vaccine development and the distinctions among mRNA, vector and inactivated COVID-19 vaccine platforms.

Status: Three audited routed objective demands are consolidated in one representative card. The locally unavailable official keys are not reconstructed, and no answer option or letter is asserted.

Model solution: Whole-pathogen platform boundary: Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. **Subunit-toxoid-VLP-conjugate boundary:** Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. **Vector-mRNA-DNA boundary:** A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. **Efficacy-effectiveness-access firewall:** Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2018 and 2022 GS-III

Demand: Discuss biotechnology activity and biopharma-sector benefits in India, and explain vaccine-development principles and Indian COVID-19 vaccine approaches.

Status: Two verified routed Mains demands are consolidated in one card. The route covers mechanism, platform choice, institutions, trial stages, manufacturing and access without inventing a UPSC model answer.

Model solution: Active-passive-antiviral boundary: Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. **Whole-pathogen platform boundary:** Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. **Vector-mRNA-DNA boundary:** A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. **Biopharma-manufacturing chain:** Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. **Clinical-development ladder:** Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use.

Phase labels do not substitute for protocol details. **CDSCO-NDCTR boundary:** CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. **DBT-BIRAC-ICMR boundary:** DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. **Mission-and-policy status boundary:** Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. **Efficacy-effectiveness-access firewall:** Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2025 Prelims GS-I

Demand: Assess the routed statements concerning monoclonal antibodies.

Status: Representative audited card for 2025 Q48. The official Set-A key is locally available in the repository, but no answer, option or letter is recorded or inferred in this authoring file.

Model solution: Monoclonal-antibody mechanism: A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. **Hybridoma-recombinant-production boundary:** Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. **Antibody-engineering boundary:** Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. **Biosimilar-generic boundary:** A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. **Efficacy-effectiveness-access firewall:** Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish antigen, antibody, active immunisation, passive immunisation and antiviral action. Answer in about 150 words.

Model thesis: Claim: Antigen-antibody boundary. **Named evidence/example:** An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Active-passive-antiviral boundary. **Named evidence/example:** Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Adaptive-immune mechanism. **Named evidence/example:** Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms.
- Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication.
- Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection.

Qualified conclusion: Claim: Antigen-antibody boundary. **Named evidence/example:** An antigen is a molecular structure recognised by an antibody or T-cell receptor, while an antibody is a B-cell-lineage immunoglobulin that binds a specific epitope; antigen, immunogen, antibody and pathogen are related but non-interchangeable terms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Active-passive-antiviral boundary. **Named evidence/example:** Vaccination presents antigen or antigen instructions so the recipient develops active immunity and memory, whereas a monoclonal antibody supplies a ready-made targeted protein for passive prophylaxis or therapy; an antiviral instead inhibits a step of viral replication. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Adaptive-immune mechanism. **Named evidence/example:** Antigen presentation activates appropriate lymphocytes; B cells can become antibody-secreting plasma cells and memory B cells, while helper and cytotoxic T-cell functions support or execute cellular responses. Antibody titre alone is not the whole immune response or a universal correlate of protection. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Compare live-attenuated, inactivated, subunit, toxoid, VLP and conjugate vaccines. Answer in about 150 words.

Model thesis: Claim: Whole-pathogen platform boundary. **Named evidence/example:** Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subunit-toxoid-VLP-conjugate boundary. **Named evidence/example:** Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Adjuvant-and-formulation boundary. **Named evidence/example:** An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical.
- Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory.
- An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration.

Qualified conclusion: Claim: Whole-pathogen platform boundary. **Named evidence/example:** Live-attenuated vaccines use a weakened replicating organism and inactivated vaccines use a killed organism; both expose multiple antigens, but attenuation, replication, contraindications, boosting and manufacture must not be treated as identical. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Subunit-toxoid-VLP-conjugate boundary. **Named evidence/example:** Subunit vaccines present selected components, toxoids present inactivated bacterial toxins, virus-like particles are non-infectious protein assemblies without a viral genome, and conjugate vaccines link a weak polysaccharide antigen to a protein carrier to strengthen T-cell-dependent response and memory. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Adjuvant-and-formulation boundary. **Named evidence/example:** An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain viral-vector, mRNA and DNA vaccine mechanisms and their delivery and cold-chain implications. Answer in about 250 words.

Model thesis: Claim: Vector-mRNA-DNA boundary. **Named evidence/example:** A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Adjuvant-and-formulation boundary. **Named evidence/example:** An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cold-chain boundary. **Named evidence/example:** Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ.
- An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration.
- Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.

Qualified conclusion: Claim: Vector-mRNA-DNA boundary. **Named evidence/example:** A recombinant viral vector delivers antigen code through an engineered carrier, mRNA is translated in the cytoplasm after delivery such as by lipid nanoparticles, and DNA must reach the nucleus for transcription; all induce active immunity but their delivery, stability and manufacturing implications differ. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Adjuvant-and-formulation boundary. **Named evidence/example:** An adjuvant enhances or shapes an immune response to antigen, whereas stabilisers, preservatives and delivery systems perform different formulation functions; the presence of an adjuvant does not itself establish efficacy, safety or duration. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cold-chain boundary. **Named evidence/example:** Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Trace monoclonal-antibody development from hybridoma or sequence discovery to recombinant manufacture and clinical use. Answer in about 250 words.

Model thesis: Claim: Monoclonal-antibody mechanism. **Named evidence/example:** A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hybridoma-recombinant-production boundary. **Named evidence/example:** Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibody-engineering boundary. **Named evidence/example:** Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biopharma-manufacturing chain. **Named evidence/example:** Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Quality-and-scale-up boundary. **Named evidence/example:** Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects.
- Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages.
- Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness.
- Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules.
- Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs.

Qualified conclusion: Claim: Monoclonal-antibody mechanism. **Named evidence/example:** A monoclonal antibody is selected for binding to one epitope or defined target and can neutralise, block signalling, recruit immune effector functions or carry a payload; targeted binding does not guarantee clinical benefit or absence of adverse effects. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Hybridoma-recombinant-production boundary. **Named evidence/example:** Classical hybridoma technology fuses an antibody-producing B cell with an immortal myeloma cell to create a clone, while modern therapeutic antibodies are commonly sequence-engineered and expressed in recombinant mammalian cell culture; discovery clone, production cell line and final medicine are separate stages. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Antibody-engineering boundary. **Named evidence/example:** Chimeric, humanised and fully human antibodies describe different degrees or routes of human-sequence content intended partly to manage immunogenicity; suffix conventions are historical naming aids, not proof of mechanism, safety, approval or effectiveness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biopharma-manufacturing chain. **Named evidence/example:** Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Quality-and-scale-up boundary. **Named evidence/example:** Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Examine India's vaccine and biologics pathway through clinical stages, CDSCO regulation, DBT-BIRAC support and pharmacovigilance. Answer in about 300 words.

Model thesis: Claim: Clinical-development ladder. **Named evidence/example:** Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CDSCO-NDCTR boundary. **Named evidence/example:** CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DBT-BIRAC-ICMR boundary. **Named evidence/example:** DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Mission-and-policy status boundary. **Named evidence/example:** Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Pharmacovigilance-and-AEFI boundary. **Named evidence/example:** Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Efficacy-effectiveness-access firewall. **Named evidence/example:** Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details.
- CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery.

- DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role.
- Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access.
- Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine.
- Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source.

Qualified conclusion: Claim: Clinical-development ladder. **Named evidence/example:** Preclinical studies precede human trials; Phase I primarily examines initial safety and dose, Phase II expands dose, safety and immune or therapeutic evidence, Phase III tests benefit and safety in larger populations, and Phase IV/post-marketing surveillance follows authorised use. Phase labels do not substitute for protocol details. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** CDSCO-NDCTR boundary. **Named evidence/example:** CDSCO under the Ministry of Health and Family Welfare is India's national drug regulatory authority, with the DCGI/Central Licensing Authority operating under the Drugs and Cosmetics framework and NDCTR 2019 for new drugs and clinical trials; regulator review is distinct from research, funding, procurement and immunisation delivery. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** DBT-BIRAC-ICMR boundary. **Named evidence/example:** DBT provides biotechnology policy, mission and research support, BIRAC is the DBT-set-up Section 8 public sector translation and industry-academia interface, and ICMR is a biomedical research and evidence institution; none replaces CDSCO's market-approval role. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Mission-and-policy status boundary. **Named evidence/example:** Mission COVID Suraksha was a DBT-led, BIRAC-implemented mission-mode vaccine-development support architecture, while BioE3 places precision biotherapeutics in a wider biomanufacturing frame; a mission, policy, call or supported facility is not a product approval or evidence of current output and access. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Pharmacovigilance-and-AEFI boundary. **Named evidence/example:** Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Efficacy-effectiveness-access firewall. **Named evidence/example:** Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Evaluate India's biopharma capability through manufacturing quality, biosimilars, cold chain, affordability, access and status discipline. Answer in about 300 words.

Model thesis: Claim: Biopharma-manufacturing chain. **Named evidence/example:** Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Quality-and-scale-up boundary. **Named evidence/example:** Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cold-chain boundary. **Named evidence/example:** Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biosimilar-generic boundary. **Named evidence/example:** A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Pharmacovigilance-and-AEFI boundary. **Named evidence/example:** Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Efficacy-effectiveness-access firewall. **Named evidence/example:** Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.

Claim -> named evidence -> analysis -> qualification:

- Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules.
- Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs.
- Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact.
- A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence.

- Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine.
- Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source.

Qualified conclusion: Claim: Biopharma-manufacturing chain. **Named evidence/example:** Biologics manufacture proceeds through cell-bank and raw-material control, upstream culture or fermentation, harvest, downstream purification, formulation, sterile fill-finish, testing and batch release; process control is part of product quality because living systems generate complex and variable molecules. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Quality-and-scale-up boundary. **Named evidence/example:** Identity, purity, potency, sterility, contamination control, consistency and validated storage are distinct quality dimensions; laboratory expression, pilot production, validated commercial manufacture and released batches are separate capability rungs. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Cold-chain boundary. **Named evidence/example:** Cold chain is temperature-controlled storage and transport from manufacturer through delivery, supported by monitoring and contingency systems; a platform's nominal storage range alone does not establish last-mile integrity, coverage, equity or field impact. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Biosimilar-generic boundary. **Named evidence/example:** A generic small-molecule medicine is expected to be chemically identical to its reference and can rely heavily on bioequivalence, whereas a biosimilar is highly similar to a reference biologic and requires a stepwise comparability exercise across quality and appropriate non-clinical, clinical and pharmacovigilance evidence. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Pharmacovigilance-and-AEFI boundary. **Named evidence/example:** Pharmacovigilance detects, assesses, understands and helps prevent adverse effects or other medicine-related problems after and around use; an adverse event following immunisation is temporally associated with vaccination but is not automatically caused by the vaccine. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary. **Claim:** Efficacy-effectiveness-access firewall. **Named evidence/example:** Trial efficacy, real-world effectiveness, regulatory authorisation, manufacturing capacity, batch release, procurement, availability, affordability, uptake and population impact are separate evidence rungs; each product claim needs its own dated authoritative source. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, institution, system category, legal character and development-to-deployment status boundary.